

ITT320 INTRO TO COMPUTER SECURITY

Chapter 5 : Encryption

Zulazeze Sahri, UiTM



Objectives

- Explain the basics of encryption
- Discuss modern cryptography methods
- Select appropriate cryptography for your organization
- Understand the function and protocols of VPNs



Introduction

- Encryption
  - Scrambling information
  - One critical part to the security puzzle
  - Without it, all security measures are inadequate

- Cryptography
  - An art form
  - The information that is being hidden is called **plaintext** and the information that has been encrypted is called **ciphertext**.

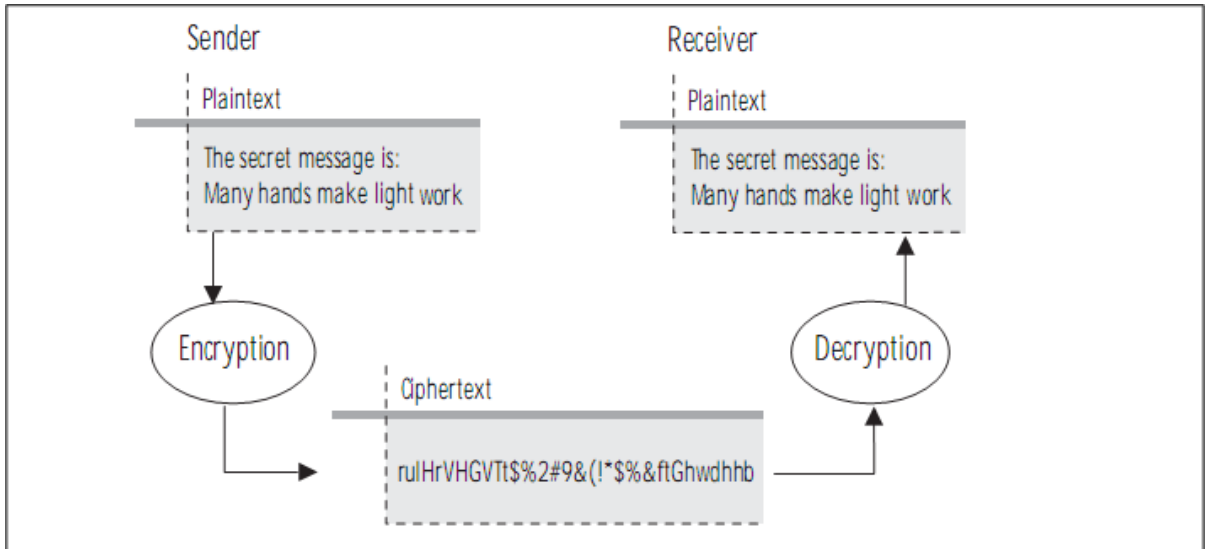


An Overview of Cryptography

- Origin from the Greek word ***kruptos*** = ***hidden***.
- **Objective**: to hide information so that only the intended recipient can unhide it.
- Hiding information = **encryption**
- Unhidden information = **decryption**
- A **cipher** is used to accomplish encryption and decryption.
- Merriam Webster's Collegiate Dictionary:
  - A cipher is **a method of transforming a text in order to conceal its meaning**.

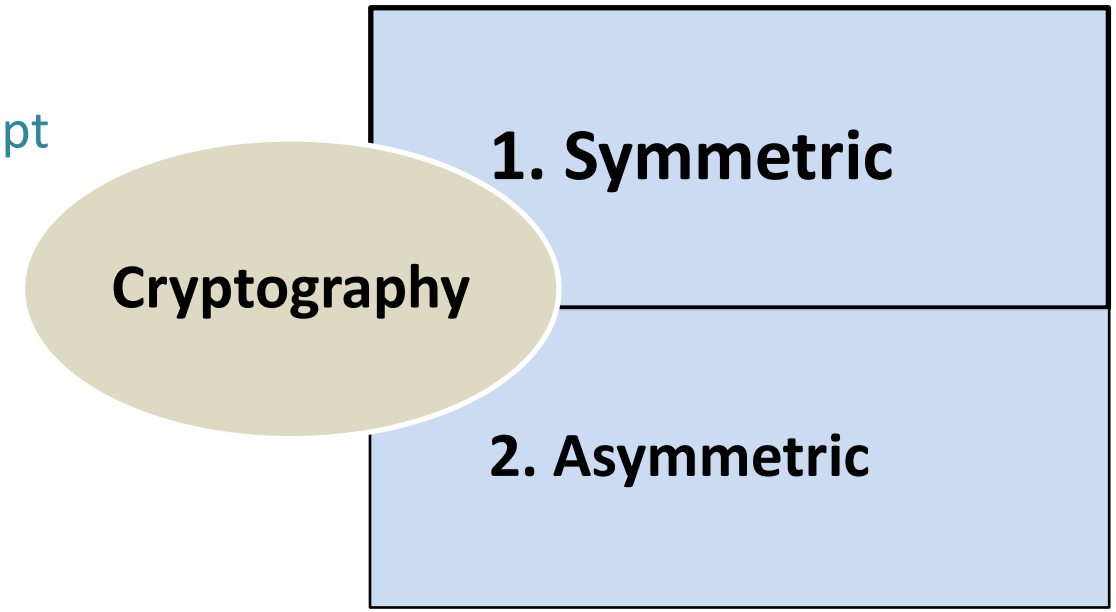
Cryptography Basics

- Decryption
  - Reversal of the scrambling protocol
- Encryption
  - Algorithm scrambles plain
  - Sender and receiver agree on algorithm
  - Message difficult to re-create without protocol



Cryptography Basics (cont.)

- Two basic types
  - Single/symmetric key encryption
    - Use a **single key** for both encryption and decryption of ciphertext
    - Single key = **secret key** because it must be kept secret from anyone
    - Stream
    - Block : Substitution and transposition
  - Public/asymmetric key encryption
    - **Two different keys** are used: one for **encryption** and one for **decryption**.
    - Also called as **public key** and **private key**
    - **Public key encryption** = **Asymmetric encryption**



Cryptography Basics (cont.)

Symmetric

use a **single key**

- **DES**
- **IDEA**
- **Blowfish**

Asymmetric

use multiple keys

- **Public Key Enc.**
- **RSA**

History of Encryption

- Old as written communication and war
- Caesar Cipher : Julius Caesar has his own cryptography called **Caesar's Cipher**
  - Rotated the letters of the alphabet to the right by three
  - Today's standards, it is extremely simplistic
  - Substitution alphabet.
    - Substitutes one letter in the alphabet for another.
    - Caesar is a mono-alphabetic cipher.

Caesar Cipher with a shift of 3

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Shift: 0

Plain Text: Hello Cipher Text:

History of Encryption (cont.)

- Multi-alphabetic
  - Select multiple shifts
    - Shift 1, 2, –1
    - Rotate through the shifts
    - A DOG becomes B FNH
  - Old cipher considered weak today



Binary Operations

- Binary Operations
  - AND, OR, XOR

- Example of AND operation

$$\begin{array}{cccc} 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ \hline 1 & 0 & 0 & 1 \end{array}$$

Binary Operations

- Example of OR operation

$$\begin{array}{r} 1\ 1\ 0\ 1 \\ 1\ 0\ 0\ 1 \\ \hline 1\ 1\ 0\ 1 \end{array}$$



Binary Operations

- Example of XOR operation

$$
\begin{array}{cccc}
1 & 1 & 0 & 1 \\
1 & 0 & 0 & 1 \\
\hline
0 & 1 & 0 & 0
\end{array}
$$



Binary Operations

- XOR only reversible binary operation
  - Convert plain text to ASCII

    A DOG = 065 032 068 079 071

  - Then, convert ASCII to binary

    0100 0001, 0100 0100, 0100 1111, 0100 0111



Binary Operations

- XOR the ASCII

  0100 0001, 0100 0100, 0100 1111, 0100 0111
  1111 0111, 1111 0111, 1111 0111, 1111 0111
  1011 0110, 1011 0011, 1011 1000, 1011 0000

- Result is cipher text.



Cryptography Terms

- Key: The bits that are combined with the plain text to encrypt it. In some cases this is random numbers; in other cases it is the result of some mathematical operation.
- Plain text: The unencrypted text.
- Cipher text: The encrypted text.
- Algorithm: A mathematical process for doing something.



Modern Methods

- Single key (symmetric) encryption
  - Same key to encrypt and decrypt
    - Blowfish
      - Symmetric block cipher
      - Works on “blocks” of letters
      - Uses variable length key (32–448 bits)
      - Freeware



Modern Methods (cont.)

- Data Encryption Standard (DES)

1. Divided into 64-bit blocks; then transposed
2. Manipulated by 16 steps of encryption, using 56-bit key
3. Scrambled by a swapping algorithm
4. Transposed one final time



Modern Methods (cont.)

- Advanced Encryption Standard (AES)

- Advanced Encryption Standard was the algorithm eventually chosen to replace DES.
- It is a block cipher that works on 128-bit blocks.
- It can have one of three key sizes of 128, 192, or 256 bits.
- This was selected by the United States government to be the replacement for DES and is now the most widely used symmetric key algorithm.



Modern Methods (cont.)

- Public key (asymmetric) encryption
  - Opposite of single key encryption.
  - One key (public key) used to encrypt .
  - One key (private key) used to decrypt.
  - Only holder of a private key can decrypt messages.



Modern Methods (cont.)

- Public key (asymmetric) encryption
  - Depends on large prime numbers, factoring, and number theory.
  - Public key encryption is most widely used.
  - Pretty Good Privacy (PGP):
    - Freeware
    - Quite secure



Modern Methods (cont.)

- Public key (asymmetric) encryption
  - Pretty Good Privacy (PGP)
    - Often used for signing, encrypting and decrypting texts, e-mails, files, directories and whole disk partitions to increase the security of e-mail communications
    - Public key encryption, Considered quite secure, free
    - Similar product : OpenPGP
    - Freeware
    - Phil Zimmerman – 2004
    - Quite secure

Modern Methods (cont.)



Welcome to the MIT Distribution Center for PGP (Pretty Good Privacy)

FLASH: PGP Freeware v6.5.8 is now available for Windows 95/98/NT/2000! and the Macintosh
PGP Freeware v6.5.8 is MacOS 7.6.1+
PGP Command Line Freeware v6.5.8 is now available for AIX/HP-UX/Linux/Solaris!
PGP Certificate Server Freeware v2.5.8 is now available for Windows NT/2000 and Solaris!

PGP® or Pretty Good Privacy® is a powerful cryptographic product family that enables people to securely exchange messages, and to secure files, disk volumes and network connections with both *privacy* and *strong authentication*.

The MIT Distribution Center for PGP home page (http://web.mit.edu/network/pgp.html)



Modern Methods (cont.)

- Public key (asymmetric) encryption
  - RSA
    - You start by generating two large random primes, p and q, of approximately equal size. Now you need to pick two numbers so that when multiplied together the product will be the size you want (that is, 128 bits, 256 bits, and so on).
    - Now multiply p and q to get n.
    - Let n = pq
    - Let m = (p - 1)(q – 1)



Modern Methods (cont.)

- Public key (asymmetric) encryption
  - RSA
    - Now select another number; call this number e. Pick e so that it is co-prime to m.
    - Choose a small number e, co-prime to m.
    - Youare almost done generating a key. Now you just find a number d that when multiplied by e and modulo m would yield a 1. (Note: Modulo means to divide two numbers and return the remainder. For example 8 modulo 3 would be 2.).
    - Find d, such that de % m = 1.
    - Now publish e and n as the public key. Keep d and n as the secret key. To encrypt, simply take your message raised to the e power and modulo n.

Modern Methods (cont.)



The RSA Security home page (http://www.rsasecurity.com)

Modern Methods (cont.)

- Legitimate versus fraudulent encryption

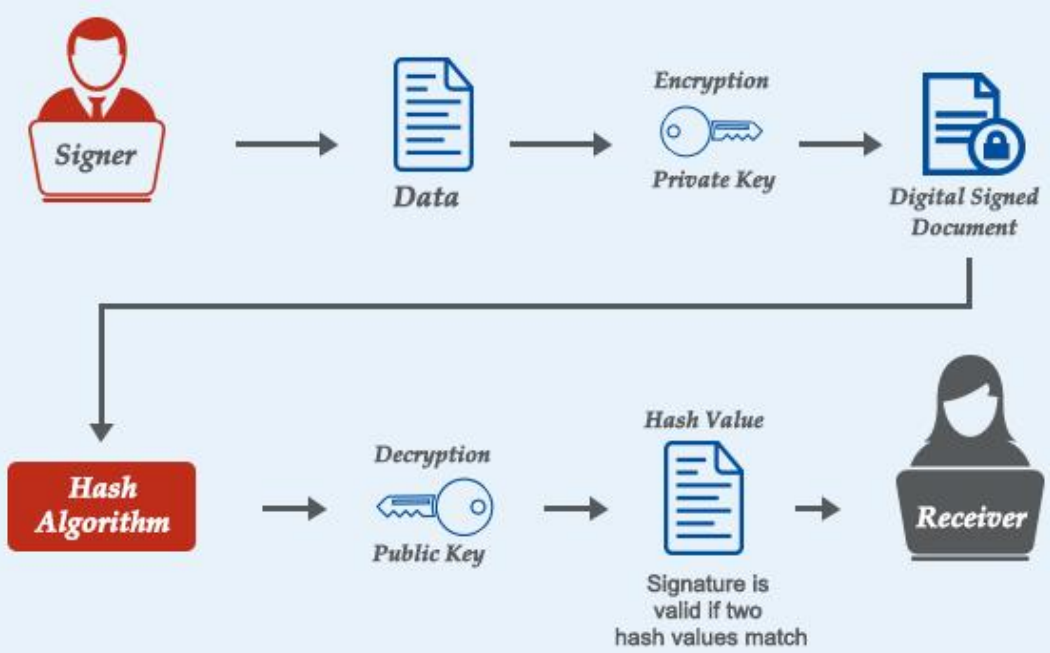
Warning signs of frauds

- Unbreakable
- Certified
- Inexperienced people



Digital Signatures

- A digital signature is not used to ensure the confidentiality of a message, but rather to guarantee who sent the message. This is referred to as nonrepudiation. Essentially, it proves who the sender is.
- Digital signatures are actually rather simple, but clever. They simply reverse the asymmetric encryption process.
- Recall that in asymmetric encryption the public key (which anyone can have access to) is used to encrypt a message to the recipient, and the private key (which is kept secure and private) can decrypt it.
- With a digital signature, the sender encrypts something with his private key. If the recipient can decrypt that with the sender's public key, then it must have been sent by the person purported to have sent the message.



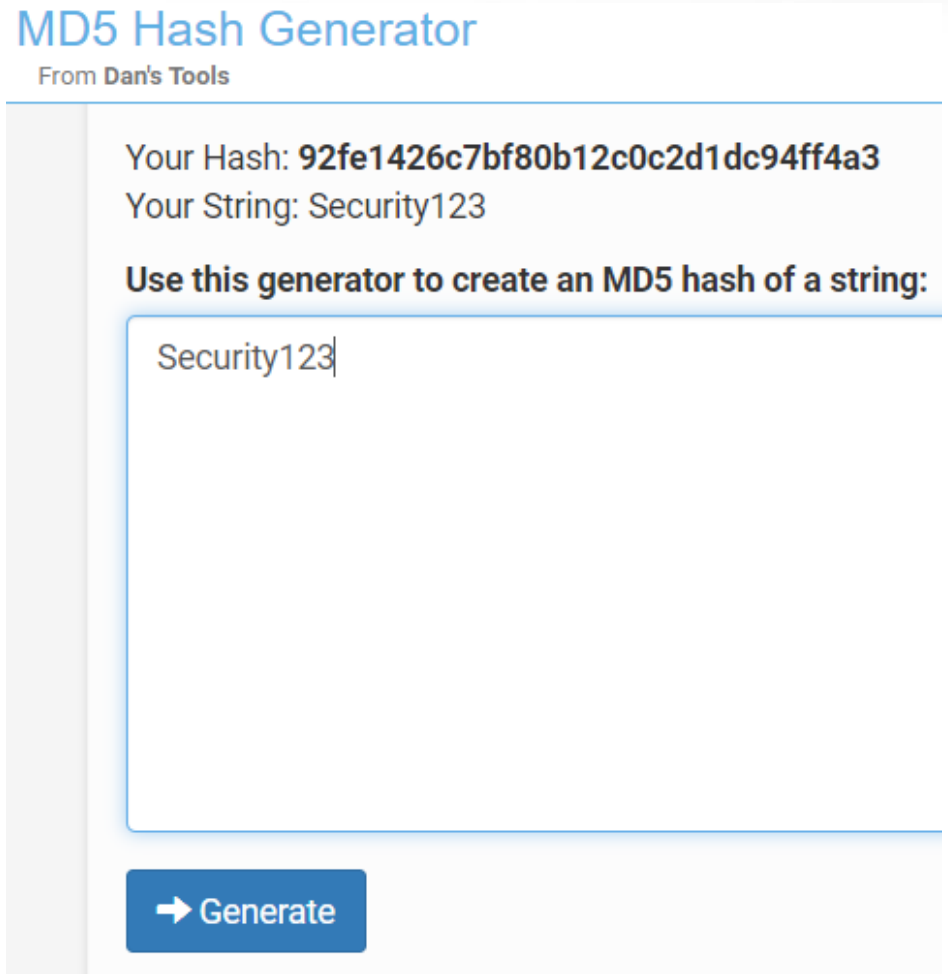


Hash

- Hashing is a type of cryptographic algorithm that has some specific characteristics. First and foremost it is one way. That means you cannot "unhash" something. The second characteristic is that you get a fixed-length output no matter what input is given. Finally, it should have few or no collisions. A collision is when two different inputs provide the same output.

Hash – cont.

- MD5 - The **MD5** message-digest algorithm is a widely used hash function producing a 128-bit hash value. Suffer from Vulnerabilities.
- SHA1 - SHA-1 (Secure Hash Algorithm 1) is a cryptographic hash function which takes an input and produces a 160-bit (20-byte) hash value
- SHA2 - Often called the **SHA-2** family of hashes because it contains many different-size hashes, including 224-, 256-, 384-, and 512-bit digests.
- SHA3 – (Secure Hash Algorithm 3) is the latest member of the Secure Hash Algorithm family of standards, released by NIST on August 5, 2015.





VERY Simple Illustration of Rainbow Tables

Password	MD5 Hash (in Hex)
aaaa	74b87337454200d4d33f80c4663dc5e5
aaab	4c189b020ceb022e0ecc42482802e2b8
aaac	3963a2ba65ac8eb1c6e2140460031925
aaa1	39dc4f1ee693e5adabddd872247e451f
aaa2	0ad346c93c16e85e2cb117ff1fcfada3
aaa4	ee93fca7c150d9c548aff721c87d0986

Password	MD5 Hash (in Hex)
aaaaa	594f803b380a41396ed63dca39503542
aaabb	120858a7016efcfab66967b834e9153c
aaacc	ee43671d755ac457cfe6e32d1894788e
aaa1a	5bbac29650eb36b4de16885c190a9fa3
aaa2a	597f0ce6d11567cc691b3f5df35594cb
aaa4a	4305dc076b3ba2bf8d55524cddf5a72d



Hash - Salt

Random bits added to further secure encryption or hashing. Most often encountered with hashing, to prevent Rainbow Table attacks.

Essentially the salt is intermixed with the message that is to be hashed. Consider this example. You have a password that is pass001

- In binary that is 01110000 01100001 01110011 01110011 00110000 00110000 00110001
- A salt algorithm would insert bits periodically, lets assume for our example that we insert bits every 4th bit giving us 0111100001 0110100011 0111100111 0111100111 0011100001 0011100001 0011100011
- If you convert that to text you would get xZ7??#



Historical Steganography

- The ancient Chinese wrapped notes in wax and swallowed them for transport.
- In ancient Greece a messenger's head might be shaved, a message written on his head, then his hair was allowed to grow back.
- In 1518 Johannes Trithmeus wrote a book on cryptography and described a technique where a message was hidden by having each letter taken as a word from a specific column.

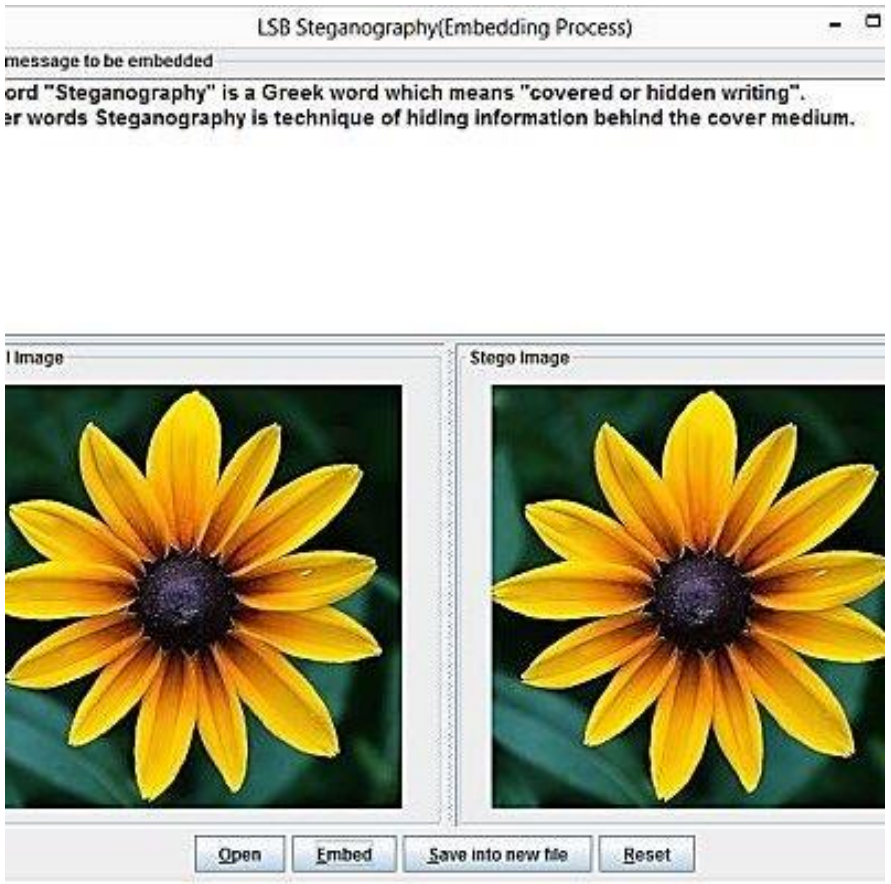


Historical Steganography – Continued

- During WW II the French Resistance sent messages written on the backs of couriers using invisible ink
- Microdots are images/undeveloped film the size of a typewriter period, embedded on an innocuous documents. These were said to be used by spy's during the Cold War.

Steganography Terms

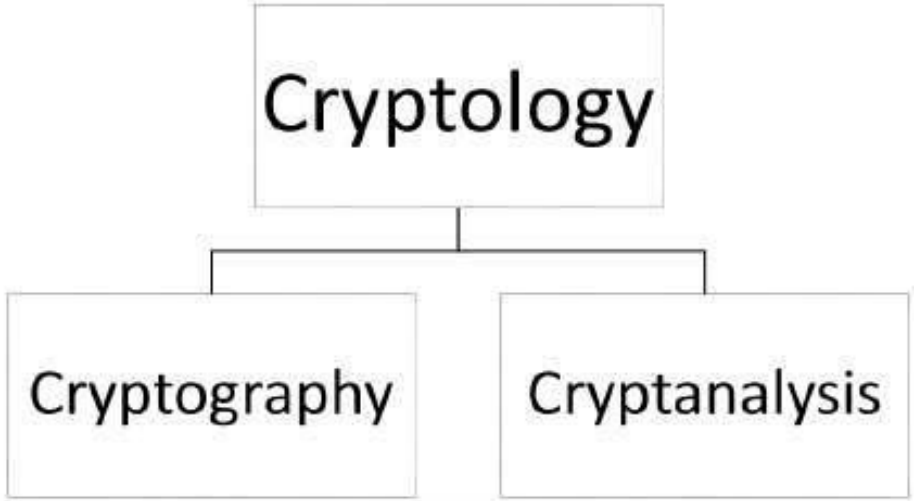
- Payload is the data to be covertly communicated.
- The carrier is the signal, stream, or data file into which the payload is hidden. This is also sometimes called the cover object.
- The channel is the type of medium used.



Cryptanalysis

Cryptanalysis is the study of ciphertext, ciphers and cryptosystems with the aim of understanding how they work and finding and improving techniques for defeating or weakening them.

- Ciphertext Only Attack
- Known plain text
- Chosen plain text
- Related Key





Summary

- Encryption is a basic element of security.
- Encrypting data when transmitting is an integral part of any security plan.